Question-1)

Answer the following questions:

a) What is the acting rationally?

b) What is thinking rationally?

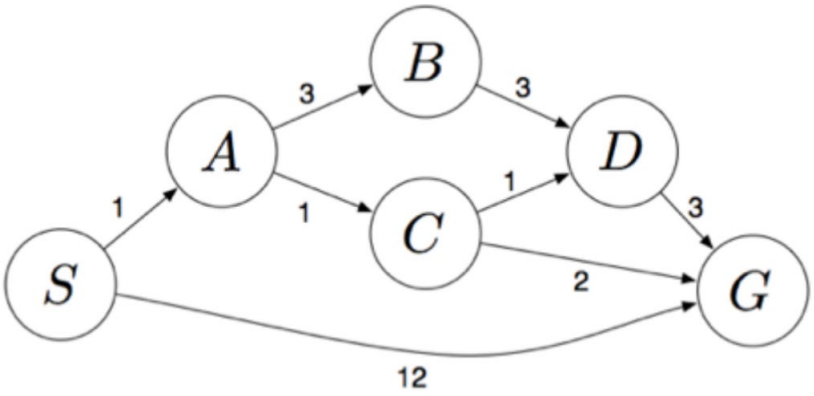
c) How can an agent act rationally without thinking rational?

Question-2)

An agent might be placed into an environment that is fully observable or partially observable, deterministic or stochastic, single-player or multi-player, episodic or sequential. The adjectives "partially observable," "stochastic," "episodic," and "single-agent" are used to describe different types of gaps in the agent's knowledge. Define each of these four adjectives: specifically, what type of knowledge about the environment is lacking, in each of these four cases? (8 points)

Question -3)

Consider the search problem with the following state space:



S denotes the start state, G denotes the goal state, and step costs are written next to each arc. Assume that ties are broken alphabetically (i.e., if there are two states with equal priority on the frontier, the state that comes first alphabetically should be visited first).

a) What path would BFS return for this problem? Explain.

b) What path would DFS return for this problem? Explain.

c) In general, discuss the relative strengths and weaknesses of breadth-first search (BFS) vs. depth-first search (DFS) for AI problems.

	BFS	DFS
Strength		
Weakness		

Question -4)

a) Convert the following sentences to first-order logic:

a) All dogs have tails

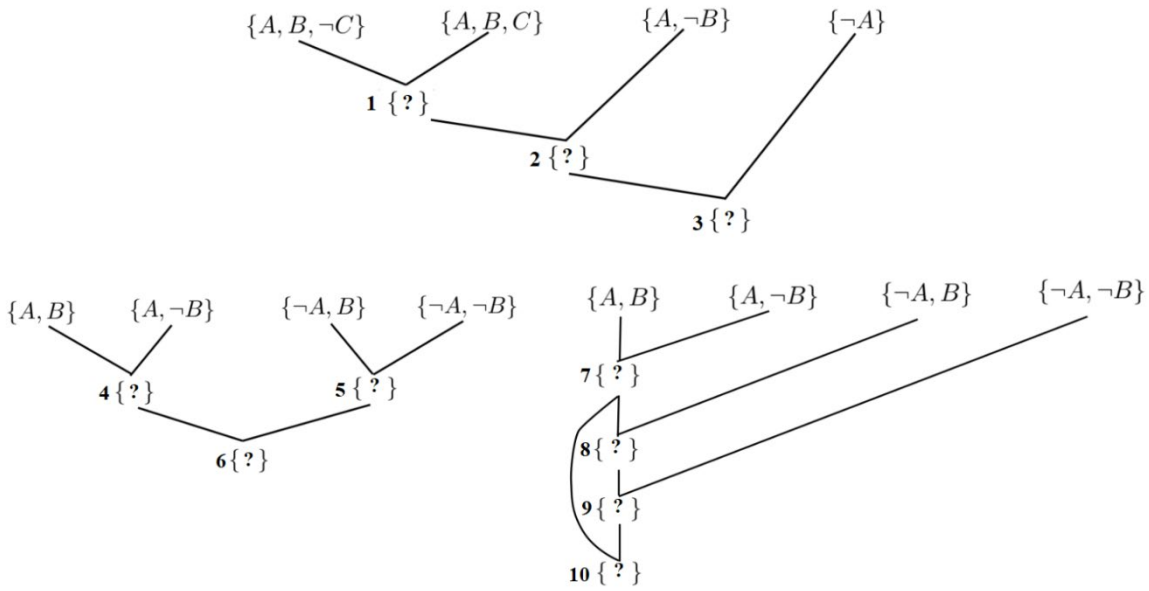
b) John is a student

c) Every dog chases some rabbit

d) Everyone who loves Santa loves any reindeer

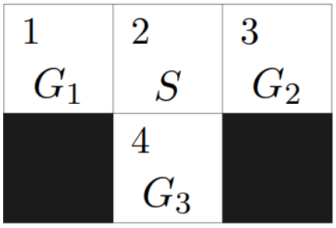
e) Anyone who wears maroon-and-white shirts is an Aggie

b) What are the answers for the question marks in the figures below? In addition, describe how you achieved answers for the question in 1, 5, and 10 shown in the figure below briefly.



Question – 5)

Imagine a maze with only four possible positions, numbered 1 through 4 in the following diagram. Position 2 is the start position (denoted S in the diagram below), while positions 1, 3, and 4 each contain a goal (denoted as G1, G2, and G3 in the diagram below). Search terminates when the agent finds a path that reaches all three goals, using the smallest possible number of steps



a) Define a notation for the state of this agent. How many distinct non-terminal states are there? (6 points)

b) Draw a search tree out to a depth of 3 moves, including repeated states. Circle repeated states. (10 points)

Question-6)

a) What are the differences between Supervised and Unsupervised learning (write down only 3).

b) Draw an Artificial Neural Network model with a single perceptron including 4 inputs and 1 output.